

Human Ecology

*To my grandchildren
Kylie, Jesse and Max*

Human Ecology

BASIC CONCEPTS FOR SUSTAINABLE DEVELOPMENT

Gerald G Marten

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Contents

<i>List of figures and boxes</i>	<i>viii</i>
<i>Foreword by Maurice Strong</i>	<i>xii</i>
<i>Preface</i>	<i>xiv</i>
<i>Acknowledgements</i>	<i>xvii</i>
1 Introduction	1
What is human ecology?	1
Sustainable development	7
Organization of this book	11
2 Populations and feedback systems	14
Exponential population growth	14
Positive feedback	17
Negative feedback	20
Population regulation	21
The practical significance of positive and negative feedback	24
3 Human population	26
Human population history	27
Social mechanisms of population regulation	35
The population explosion and quality of life	38
4 Ecosystems and social systems as complex adaptive systems	42
Hierarchical organization and emergent properties	43
Self-organization	47
Stability domains	52
Complex system cycles	56
5 Ecosystem organization	60
Coadaptation	61
Ecosystem design	62
Ecosystem homeostasis	63

vi *Human Ecology*

Comparison of natural, agricultural and urban ecosystems	65
Landscape mosaics	71
6 Ecological succession	76
Ecological succession	77
Human-induced succession	83
Managing succession	88
7 Coevolution and coadaptation of human social systems and ecosystems	96
Coadaptation in traditional social systems	97
Coevolution of the social system and ecosystem from traditional to modern agriculture	100
8 Ecosystem services	106
Material cycling and energy flow	107
Ecosystem services	114
The relation between ecosystem services and intensity of use	115
The fallacy that economic supply and demand protect natural resources from overexploitation	118
9 Perceptions of nature	121
Common perceptions of nature	122
Attitudes of religions toward nature	125
Notes of caution about romanticizing nature and traditional social systems	133
10 Unsustainable human-ecosystem interaction	136
Human migrations	137
New technologies	141
Portable capital in a free market economy	142
Tragedy of the commons	143
Large inputs to agricultural and urban ecosystems	145
Urbanization and alienation from nature	146
The rise and fall of complex societies	147
Wishful thinking and the precautionary principle	152
11 Sustainable human-ecosystem interaction	157
Human social institutions and sustainable use of common property resources	158
Coexistence of urban ecosystems with nature	165
Resilience and sustainable development	169
Adaptive development	174

12 Examples of ecologically sustainable development	183
Dengue hemorrhagic fever, mosquitoes and copepods: an example of eco-technology for sustainable development	184
The Barataria-Terrebonne National Estuary Program: an example of regional environmental management	196
<i>Glossary</i>	215
<i>Further reading</i>	224
<i>Index</i>	229

List of Figures and Boxes

FIGURES

1.1	Interaction of the human social system with the ecosystem	2
1.2	Chain of effects through ecosystem and social system (commercial fishing in the ocean)	3
1.3	Long line fishing	4
1.4	Deforestation and cooking fuel (chain of effects through ecosystem and social system)	5
1.5	Chain of effects through social system and ecosystem when biofuel generators are introduced to villages	6
1.6	Human use of natural resources	10
2.1	Growth in the water hyacinth population during the first four months after introducing a single hyacinth to a lake	15
2.2	Water hyacinth population after 30 months	15
2.3	Exponential growth of the water hyacinth population	16
2.4	A positive feedback loop	17
2.5	The positive feedback loop that generates exponential population growth	17
2.6	Positive feedback loops that caused VHS to replace BetaMax	19
2.7	Control of body temperature by negative feedback	20
2.8	Population regulation by food supply	22
2.9	The relation of population change to carrying capacity	22
2.10	Growth of a deer population after starting with one pair of deer	23
2.11	The sigmoid curve for population growth and regulation	24
3.1	Distribution of biological production among plants and animals in the ecosystem food web	27
3.2	Growth and geographic distribution of the human population during the past 2000 years	29

List of Figures and Boxes ix

3.3	Periodic increases in human population and carrying capacity	30
3.4	Human effort required for technologies that provide higher food production	30
3.5	Ten thousand years of human population growth	31
3.6	Increase in the populations of industrialized and developing world nations from 1800–2000	32
3.7	Increase in carrying capacity and human population since the Industrial Revolution	33
3.8	Approximate percentage of the Earth's land-based biological production controlled by humans	34
3.9	Negative feedback loops for social control of human population	36
4.1	Initial food chain in the island story	48
4.2	Final food web in the island story	49
4.3	Body state of 'alive' and 'dead' to illustrate stability domains	53
4.4	Alternative stability domains with nails	54
4.5	The complex system cycle	56
4.6	Complex system cycles from the perspective of stability domains	57
5.1	Change in daisy populations that keeps the temperature of Daisy World the same over a broad range of sunlight intensities	66
5.2	Input–output exchanges of materials, energy and information between adjacent ecosystems	67
5.3	Inputs and outputs of materials, energy and information with agricultural and urban ecosystems	68
5.4	Inputs and outputs of materials, energy and information with urban ecosystems	70
5.5	Expected growth of human population in cities during the next 20 years	70
5.6	Map view of a typical landscape mosaic in western Japan	71
5.7	Typical landscape profile in western Japan	72
5.8	Spatial hierarchy of ecosystems in an urban metropolitan area	73
6.1	Typical ecological succession on deep soil in western Japan	79
6.2	Ecological succession in western Japan	80
6.3	Ecological succession as a complex system cycle	81
6.4	Competition between shrubs and grasses for sunlight and water	82
6.5	Competition between grasses that are nutritious or not nutritious for sunlight, water and mineral nutrients	84
6.6	Human-induced succession from grassland to desert caused by overgrazing	86
6.7	Disappearance of commercial fish due to overfishing	87
6.8	The okay/not okay principle of human-induced succession	88
6.9	Natural regulation of forest litter by fire (no fire protection) and accumulation of litter with fire protection	90
6.10	Landscape mosaic of a forest without fire protection	91
6.11	Succession of a Javanese polyculture field from domination by annual field crops to domination by tree crops	94

X Human Ecology

7.1 Interaction, coevolution and coadaptation of the human social system with the ecosystem	97
7.2 Traditional house design of mountain people and lowland people in Vietnam	98
7.3 Interaction of the social system with agricultural ecosystems after the Industrial Revolution	102
7.4 Coadaptation of modern social systems and ecosystems	104
8.1 Human use of renewable resources and return to the ecosystem as waste	106
8.2 Material cycling and energy flow through the ecosystem	108
8.3 The second law of thermodynamics: conversion of energy to heat when energy is transformed from one form to another	110
8.4 Flow of energy through an ecosystem food web	110
8.5 A pot of water as a metaphor for material cycling and energy flow in ecosystems	111
8.6 Energy flow from one step of a food chain to another	112
8.7 Energy flow through an entire food chain	113
8.8 Energy input–output of planet Earth	113
8.9 Ecosystem services as material cycling in the ecosystem	114
8.10 The relation between ecosystem services and intensity of use	115
8.11 Disappearance of an ecosystem service due to overexploitation	117
8.12 The control of resource use by supply and demand	119
9.1 The role of perceptions of nature in decisions affecting ecosystems	122
9.2 Stability domain diagrams for different perceptions of nature	124
10.1 Comparison of the natural Great Plains ecosystem (perennial grasses) with annual crops planted by Europeans	138
10.2 Great Plains food chain for people before and after the European invasion of North America	139
10.3 The response of fish catches to fishing intensity as an example of tragedy of the commons	144
10.4 Positive feedback loop between social complexity and the growth of cities	148
10.5 Benefits and costs of social complexity	150
10.6 Positive feedback loops for the dissolution phase in the rise and fall of complex societies	151
10.7 Change from one stability domain to another when fishing is too close to the boundary between stability domains in a fisheries ecosystem with natural climatic fluctuations	153
11.1 A system used to prevent tragedy of the commons in a coastal fishery in Turkey	161
11.2 Landscape mosaics of urban and natural ecosystems	166
11.3 Stability domain diagrams comparing high and low resilience	169
12.1 <i>Mesocyclops</i>	190
12.2 Barataria-Terrebonne estuary	197
12.3 Landscape profile showing the three major types of natural ecosystems in the estuary: swamp, marsh and open water	198

List of Figures and Boxes xi

12.4 Change in topography of Barataria-Terrebonne estuary during the past 130 years and projected 20 years into the future by a computer model	201
12.5 A 110-square kilometre section of the estuary that experienced dramatic transformation of marsh to open water during the past 50 years (Leeville, Louisiana)	202

BOXES

1.1 Intensity of demands on ecosystems	11
11.1 Examples of environments and their characteristics	180
12.1 First Barataria-Terrebonne estuary planning workshop: vision statement	205
12.2 Second Barataria-Terrebonne estuary planning workshop: obstacles and challenges	206
12.3 Third Barataria-Terrebonne estuary planning workshop: challenges and actions	207
12.4 The Barataria-Terrebonne National Estuary Program environmental management plan	208

Foreword

Ecologically sustainable development has become a universal concern. It is a challenge that merits the attention and action of us all. Efforts to promote sustainable development are proceeding on a variety of fronts, but it is still far from a reality.

A crucial ingredient for sustainable development is a well-informed public. All actions that impact the environment come ultimately from individuals. It is public opinion that stimulates governments, corporations and other sectors of society to appropriate action. Even political leaders who are strongly committed to sustainable development cannot impose it upon people who do not understand or appreciate its importance. Conversely, even the most reluctant of political leaders cannot fail to pursue sustainability when their people demand it.

Most people are concerned about the environment but feel overwhelmed by the complexity and scale of the problems. Given the diversity of competing perspectives and interests, it is often difficult to know what information to trust. The numerous forces, both social and ecological, that stand in the way of sustainable development have so much momentum that changing the present course of environmental deterioration seems unlikely without major changes in people's attitudes and behavior. The imperative of sustainable development is forcing us to think in new ways, but the way to an ecologically sustainable future is not at all clear.

Ecologically sustainable development may only be possible once we have grasped the fundamental interdependence of human society and the natural environment. Human ecology, as the science of human–environment interaction, provides a whole-system perspective that bridges the gap between the natural and social sciences. It is a broad perspective that can help to clarify environmental issues and suggest how to deal with them. While human ecology has proved its worth as an interdisciplinary approach to solving environmental problems, it has not yet attained a clear identity with an established body of theory. The time has come for human ecology to become a major scientific discipline in its own right. The stake that we all have in an ecologically healthy future is far too great to settle for less.

Human Ecology: Basic Concepts for Sustainable Development is a valuable step towards making human ecology a scientific discipline which everyone can and should understand as a guide to their own actions. Dr Marten presents a coherent

set of concepts about how ecosystems function and how human social systems interact with ecosystems. This will help readers to make sense of the complexity of human–environment interactions, enabling them to make connections not previously noticed in many areas of their lives. The examples that Dr Marten uses to illustrate the concepts are drawn from actual situations that cover a wide range of topics and take us to diverse parts of the world. They allow us to see how to make the jump from theory to practice.

For example, once aware of ‘complex adaptive systems’, readers will begin to notice ordered patterns in the changes occurring all around them. Once aware of ‘landscape mosaics’, readers will pay more attention to the implications of changes in their landscape for the future quality of their lives. The ‘switch’ character of ecosystem function clarifies how inappropriate use of environmental resources can lead to irreversible degradation. The concept of ‘coadaptation’ between social systems and ecosystems helps to explain why modern society has environmental problems, while pointing to fundamental changes necessary to deal with the problems.

Dr Marten provides us with the conceptual tools to understand and evaluate the complexities we face, so that we are better equipped to choose actions with positive outcomes in both the short and long term. The ultimate pay-offs can be realized when the ecological and systems concepts in this book are applied to sustainable development. The book explains how the existing economic system and other contemporary social institutions promote unsustainable human–environment interaction and describes social institutions that can contribute to ecologically sustainable interaction. It provides examples of successful actions by government, the private sector and civil society to develop healthier relationships with the environment.

The scope and clarity of this book make it accessible and informative to a wide readership. Its messages should be an essential component of the education for all students from secondary school to university. The book will be equally meaningful to anyone concerned with the environment who desires a fundamental understanding of the forces shaping the future of his children and grandchildren and all who care about the millions of people whose lives have been destroyed or undermined by environmental deterioration. In short, this is a book that provides a clear and comprehensible account of concepts that can be applied in our individual and collective lives to pursue the promising and secure future to which we all aspire.

Maurice Strong
Chairman, Earth Council
Former Secretary General, United Nations
Conference on Environment and Development

Preface

Twenty years ago I worked with a network of agronomists, social scientists and ecologists in South-East Asia. The agronomists were using Green Revolution technology to develop new production systems for small-scale agriculture in the region. Change was essential to feed a growing population. The new methods offered impressive possibilities for increasing food production and farmer incomes but in many instances the farmers were not using them. The agronomists were understandably disappointed and sought the help of social science colleagues to find out what they could do to convince farmers to take advantage of the opportunities.

The social scientists spent a lot of time talking to farmers and soon discovered that the farmers were generally more innovative than conservative. Many routinely experimented with new crops and cultivation techniques in a corner of their farm, adopting them on a larger scale when it was apparent they would do well. The farmers knew from experience that new crop varieties frequently did not perform so well on the marginal land that many of them had to use. Even if the land were suitable, improved varieties would require an ample water supply and expensive inputs such as fertilizers and pesticides in order to achieve high yields, inputs which were beyond the means of many poor farmers. The farmers were also afraid that new cultivation methods could lead to unanticipated long-term problems that they did not have the resources to deal with. Their apprehensions turned out to be justified, as soil problems appeared after several years in some areas with new agricultural systems. Most farmers felt they could not afford to make irreversible mistakes because they had no alternatives for earning a living besides their farms.

As the agronomists and social scientists learned more from the farmers, they came to appreciate how important it was for the new agricultural technology to fit into the everyday realities of the farmers' lives and how important it was for it to be ecologically sustainable. They also discovered that they had much to learn from the farmers' traditional agricultural methods, which had proved ecologically sustainable for centuries. All of this made the agronomists realize they would be more successful working with a broader perspective than in the past. The result was a collaborative research network of agronomists, social scientists, ecologists and farmers, each of which had something to contribute to the task of developing agriculture that farmers really could use.

This story illustrates just one of the thousands of challenges facing humanity as economies, technologies, cultures and the ways that people use natural resources vary in today's rapidly changing world. These challenges are encapsulated by the concept of 'sustainable development', frequently defined as meeting the needs of the present without compromising the ability of future generations to meet their own needs. Some consider sustainable development to be the major challenge of our time, but it is difficult for concerned individuals to know what they can actually do to help make it happen.

Human ecology, the science of interrelationships between people and the environment, can help to sharpen our perceptions of what is happening with the environment, and how people and the environment function together. This kind of understanding is one essential step toward effective action.

The term 'human ecology' has a long and varied history. It was first used in the 1920s, when a small group of urban sociologists used ecological concepts to explain what they were observing in cities. The sociologists found ecological metaphors to be of use because some of the early ecological concepts were in fact general systems concepts that were also valid for human society. This form of human ecology, which was prominent until the 1970s, was totally different from the human ecology in this book.

In the 1960s and 1970s, biological ecologists who were alarmed by the human population explosion and its implications for environmental destruction used the term 'human ecology' to emphasize that humans are subject to the same ecological limitations as other animals. At the same time, anthropologists directed attention to the shaping of culture by environment, and some anthropologists initiated human ecology field studies using prominent ecological concepts of the time, such as population regulation and energy flow. As awareness of environmental problems increased during the 1970s, scholars in a diversity of academic disciplines began to speak of 'human ecology'. These different forms of human ecology, conforming to the different disciplines in which they arose, generally had little in common with one another beyond dealing with people and the environment.

By the 1980s, biological ecologists and social scientists were working together in multidisciplinary research teams addressing practical problems involving the environment. For many of these scientists, including myself, human ecology was a perspective for problem solving that focused on interactions between human societies and the environment. By tracing chains of effects through ecosystems and human society, and by understanding more generally how people interact with ecosystems, human ecology could help to:

- anticipate the long-range environmental consequences of human actions;
- avoid disastrous surprises from the environment;
- generate ideas for dealing with environmental problems; and, in general,
- maintain a liveable and sustainable relationship with the environment.

The number of scientists studying human–environment interaction increased over the years, and the diversity of their backgrounds and perspectives expanded correspondingly.

xvi *Human Ecology*

This book grew out of an undergraduate course in the School of Policy Studies at Kwansei Gakuin University in Japan. The purpose of the course was to offer students a conceptual foundation for making human ecology and sustainable development an integral part of their personal and professional lives. The book reflects my own view of human ecology and attempts to provide a clear, understandable and coherent system of concepts for comprehending how ecosystems function and how human societies interact with ecosystems. The book draws on a variety of existing ideas, integrating long-standing ecological principles with concepts assembled more recently under the rubric of complex systems theory. The last part of the book focuses on social processes, social institutions and technologies that conflict with, or contribute to, ecologically sustainable development. While I hope this book will see extensive use in the classroom, it will be of equal value to anyone who is concerned about the environment and seeks the kind of understanding that can help to do things that make a positive difference.

Most of the concepts in this book are firmly established but a few are still the subject of unresolved scientific debates. One example is the concept of emergent properties (introduced in Chapter 4). Some scientists, including myself, consider each level of biological organization to have 'a life of its own' that emerges from the organization of the parts rather than simply from the characteristics of the parts themselves. Other scientists consider emergent properties to be theoretical constructs that lack rigorous scientific confirmation. I use emergent properties because they encourage the kind of whole-system thinking that I consider essential for ecologically sustainable development.

The concepts in this book are communicated by means of examples and stories, often accompanied by diagrams. I have sometimes used a 'broad brush' approach in the examples as the focus is on communicating the concepts rather than analysing particular situations in detail. The book includes a list of Further Reading which provides readers with the opportunity to explore central concepts and key examples in greater detail. It also offers an opportunity to explore details and perspectives of human ecology that are not covered in depth in this book.

A full appreciation of human ecology comes only with exploring the details of human–environment interaction in specific cases. The 'Things to Think About' exercises at the end of each chapter are intended to stimulate concrete translations of concepts to the reader's own circumstances. Reports in the annual *State of the World* and *Worldwatch Paper* series (Worldwatch Institute, Washington, DC) are valuable sources of information for some of these exercises. To realize the fullest benefits, readers should develop their own examples – their own 'stories' – in as much detail as possible. In that way, human ecology will come to life, and readers will begin to develop concrete visions of the possibilities for ecologically sustainable development in their own community.

Technical terms, as well as words used somewhat differently from their general meaning, are presented in bold type the first time they appear in the text. Their meanings are explained in the glossary.

Acknowledgements

While most of the ideas in this book derive from many years of ecological science, it is appropriate to recognize at least some of the specific sources of concepts and examples presented here. I should first note my intellectual debt to Kenneth E F Watt and C S Holling who pioneered the rigorous application of systems analysis to ecology. While the cyclical character of natural and social processes has been recognized for thousands of years, Holling's analysis of cycles in ecological systems stimulated the 'complex system cycles' in this book; the conflict between 'stability' and 'resilience' also originated with Holling. The book's central conceptual framework based on interaction between human social systems and ecosystems originated with Terry Rambo, who illustrated the concept with the example of cooking fuel and deforestation in India that is used in this book. I am particularly grateful for Terry Rambo's intellectual companionship over the years. The nail puzzle in Figure 4.4 was brought to my attention by Virginia Fine. The 'Gaia hypothesis' came from James Lovelock. Coadaptation and coevolution between the social system and ecosystem after the Industrial Revolution is based on Richard Norgaard's *Development Betrayed*. Some of the perceptions of nature in Chapter 9 are based on ideas developed by Gene Barrett. The discussion of the impact of 'portable capital' on renewable resource use is based on ideas originally put forth by Colin Clark. The phrase 'tragedy of the commons' was first used by Garret Hardin. The role of social complexity in the rise and fall of civilizations is based on Joseph Tainter's *Collapse of Complex Societies*. Conditions for sustainable common property management and the example of fisheries in Turkey are from Eleanor Ostrum's *Governing the Commons*, and the account of traditional forest management in Japan is based on publications by Margaret McKean. Scott Halstead provided information on the epidemiology of dengue hemorrhagic fever and Vu Sinh Nam provided information on his copepod programme for dengue control in Vietnam. Kerry St Pe provided information on the Barataria-Terrebonne National Estuary Program. Richard Carriere, Richard Borden, Anthony Clayton and Ann Marten commented on the entire manuscript at different stages of its development. Gary Haley and Julie Marten did computer graphics for the figures.

